# AMIT SINGH RAJAWAT

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## EDUCATION

**Madhav Institute of Technology & Science, Gwalior** — CGPA: 8.65
*B.Tech in Information Technology (Artificial Intelligence & Robotics)* — *Aug. 2020 – May 2024*

## WORK EXPERIENCE

**Software Engineer** — June 2024 – Present
*Kloudspot Inc.* — *Bengaluru, India*

- Designed a **Face Recognition Algorithm** in **C++** for **1:1 Verification and 1:N Identification**, achieving a global rank of **277** in 1:1 Verification and **59** in 1:N Identification (NIST-Face Recognition Vendor Test – FRVT).
- Trained models using **Multiple Architectures** (ResNet, ResNet with CBAM, ResNet with attention, etc.), incorporating rigorous **hyperparameter tuning**, **knowledge distillation**, and **reinforcement learning** techniques to enhance model performance and improve accuracy by **13%**.
- Developed an **Face Recognition System (FRS) Analysis Tool** using clustering techniques and a **FRS Registration Tool** in **Python**—implementing detection, landmark processing, and recognition for multi-pose registration with real-time tracking and visualization.
- Engineered and maintained scalable **data pipelines** for real-time facial recognition data; integrated **CI/CD practices** for automated deployment and continuous improvement of deep learning models in production environments.
- Collaborated with product managers and analytics teams to define key performance metrics, conduct exploratory data analysis, and optimize ML solutions that drive operational efficiency and business impact.

**Software Engineering Intern** — Feb 2024 – May 2024
*Kloudspot Inc.* — *Bengaluru, India*

- Engineered a high-performance **computer vision inferencing framework** in **C++** using **GStreamer**, **OpenVINO**, and **DLStreamer**, resulting in a **9%** improvement in processing speed.
- Implemented **gRPC** and a **GStreamer RTSP Server** for efficient client-server communication and low-latency video broadcasting—aligning with real-time solution needs.
- Optimized pre-processing and post-processing workflows, increasing model prediction accuracy and system efficiency for large-scale data handling.

**Machine Learning Intern (LOR)** — Dec 2023 – Jan 2024
*Indian Space Research Organisation (ISRO)* — *Bengaluru, India*

- Developed a **vision-based satellite pose estimation system** using the **HRNet algorithm**, iterative hyperparameter tuning, and real-time processing optimizations to achieve **90%** accuracy in landmark detection across diverse and challenging imaging conditions.
- Integrated **deep landmark regression** and **nonlinear pose refinement** techniques, reducing error rates by **15%** through extensive model training.

**Amazon ML Summer School** — Sept 2023 – Oct 2023
*Amazon* — *Remote*

- Implemented various **machine learning algorithms** using advanced **Python libraries**, achieving over **95%** accuracy across multiple projects.
- Worked on projects involving **sequence modeling** and distributed data processing frameworks, preparing models for high-throughput, low-latency applications.

## PROJECTS

**EchoNeuron AI Engine** | *Python, FastAPI, LangChain, ChatGroq, Tavily, JavaScript, Render* — **(Link)**

- Built **EchoNeuron AI Engine**, a unified assistant featuring live web lookup, DuckDuckGo search, ArXiv search, YouTube and PDF summarization, tutoring, and quiz generation using **FastAPI**, **LangChain**, and **ChatGroq**.
- Developed a GUI containerized with **Docker**, and implemented a secure CI/CD pipeline on **Render** with safe handling of Groq & Tavily API keys.

**StoryCraftAI - Infinite Possibilities in Every Story** | *Python, TensorFlow, Keras, NLP, Pandas, Huggingface, CI/CD* — **(Link)**

- **Developed** and **optimized** a suite of **deep learning models** for **story generation** using advanced NLP techniques. Improved model accuracies: GRU increased by **13.46%**, LSTM by **33.53%**, Bidirectional-LSTM by **25.32%**, and Bidirectional-GRU by **9.91%**.
- **Integrated** an interactive **Flask-based web interface** for **real-time story generation** and **deployed** the solution via **CI/CD** on **Hugging Face Spaces**.

**GestureSymphony - Control Video Playback Using Your Hands** | *Python, OpenCV, MediaPipe, Pygame, Flask, NumPy, Bootstrap* — **(Link)**

- Engineered a real-time **hand gesture recognition** system using **MediaPipe Hands** and **OpenCV** for detection and control, demonstrating strong **computer vision and ML** integration.
- Enhanced system efficiency by implementing **lazy loading** and containerizing with **Docker**; established a robust **CI/CD pipeline** for continuous deployment on cloud platforms like **Render**.

## PUBLICATIONS

- A. Rajawat, A. Manjhi and A. Srivastava, "**Categorization of IRIS Flower using Different Machine Learning Strategies,**" [paper].
- A. Manjhi, A. Rajawat and A. Srivastava, "**Design and Analysis of Programmed Face Monitoring System,**" [paper].
- Rajawat, A.S., Srivastava, A., "**Recognition of Parkinson's Ailment by Using Various Machine Learning Procedures**." Current Psychology Journal (2024). [paper].
- **On Heart Disease** (Communicated in Journal).

## TECHNICAL SKILLS

- **Programming Languages**: C/C++ (Proficient), Python (Proficient), JavaScript, HTML/CSS, Shell, MySQL
- **Developer Tools & Frameworks**: Git, GitHub, Linux, Docker, VS Code, ONNX, AWS, CI/CD, MLOps
- **Libraries**: Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn, Flask, OpenCV, Transformers , MXNet, LLMs
- **Coursework**: DBMS, Machine Learning and Optimization, OS, DSA, CN, Robotics, Computer Vision, NLP, AI

## ACHIEVEMENTS

- **Leetcode:** 850+ Problems Solved & 1765+ Rating **(Profile Link)**, **GFG:** 320+ Problems Solved **(Profile Link)**,
- Selected for **IIT Madras Pravartak Undergraduate Fellowship 2023-24** (among 72 in India) **(Link)**.
- Designed and developed the NEC Portal for the college, currently used by over 5,000 students **(Link)**.
- **Certifications:** Advanced Learning Algorithms , DeepLearning.AI TensorFlow Developer Specialization , Machine Learning